ThermoTwin

A First-Principles Digital Twin for Refinery Fired Heaters

Governing equations, data generation, surrogate model and optimizer

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This document states every equation used in the project, the assumptions behind them, and how the pieces fit together, so that a new reader can follow the work from the combustion chemistry to the final fuel and carbon savings.

Project	Physics + machine-learning twin for fired-heater optimization
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Source specification	`HEATER KPIs.xlsx` (roadmap, compositions, KPI formulas)
Code	`C:\trio\heater_sim`
Status	Prototype validated on first-principles simulation

All quantities use SI units unless stated. Heating values are lower heating values at 25 °C unless marked HHV.

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1 What this project does

A fired heater burns fuel gas to raise the temperature of a process stream. It is the largest single fuel user and carbon source on a refinery unit. The heater runs with extra combustion air as a safety margin against incomplete burning, and that extra air carries heat up the stack, which wastes fuel. The safe optimum moves with fuel composition, throughput, fouling and weather, so a static setting is never optimal for long.

ThermoTwin is a digital twin of the heater. It has three layers:

1. A **first-principles model** that closes the combustion and energy balance from physics alone. It needs no plant history, so it is useful on the first day of a deployment (Sections 2–5).
2. A **data generator** that drives the model through a realistic operating history with drift and labelled faults, producing a dataset that stands in for a plant historian at the ideation stage (Section 6).
3. A **machine-learning surrogate** trained on that data, plus an **optimizer** that recommends the setpoint which minimises fuel and carbon within the safety limits (Sections 7–8).

The roadmap from the source workbook is Collect, Clean, Label, Model, Train, Validate, Deploy. Since no plant historian exists yet, the Collect step is replaced by the first-principles simulation.

1.1 Notation

Symbol	Meaning	Unit
$\dot{m}_p$	process-fluid mass flow	kg/h
$c_{p,p}$	process-fluid specific heat	kJ/(kg K)
$T_{\text{in}}, T_{\text{out}}$	process inlet and outlet temperature	°C
Q	heater duty (heat delivered to the process)	MW or GJ/h
x_i	mole fraction of fuel component i	–
M_i, LHV_i	molar mass and molar heating value of i	g/mol, kJ/mol
M_f, h_L, h_H	fuel molar mass, mass LHV, mass HHV	g/mol, MJ/kg
W	Wobbe index	MJ/Nm^3
e	excess-air fraction (0.15 means 15 %)	–
AFR	stoichiometric air-to-fuel mass ratio	kg/kg
O_2^{dry}	dry flue-gas oxygen	%
T_s	stack (flue-gas) temperature	°C
η	thermal efficiency	%
ϕ	fouling index, 0 clean to 1 fouled	–

Air is modelled as 20.95 % oxygen by mole, so each mole of oxygen is accompanied by $\beta = (1 - 0.2095)/0.2095 = 3.773$ moles of nitrogen. Mean air molar mass is taken as 28.964 g/mol.

2 Fuel-gas thermochemistry

The fuel gas is a mixture of hydrogen, light hydrocarbons and inert gases. All fuel properties are derived from the composition and the per-component data in Table 1, not read from a bulk lookup.

2.1 Composition and molar mass

A raw composition is first normalised so the mole fractions sum to one:

$$x_i = \frac{\tilde{x}_i}{\sum_j \tilde{x}_j}. \tag{1}$$

The mixture molar mass and the molar heating values follow from a mole-weighted sum:

$$M_f = \sum_i x_i M_i, \qquad \mathrm{LHV_{mol}} = \sum_i x_i\, \mathrm{LHV}_i. \tag{2}$$

2.2 Higher heating value

The higher heating value adds back the latent heat of the water formed. With $n_{H,i}$ hydrogen atoms per molecule, each molecule yields $n_{H,i}/2$ moles of water, and with a latent heat $\lambda = 44\,\mathrm{kJ/mol}$:

$$\mathrm{HHV_{mol}} = \sum_i x_i \left(\mathrm{LHV}_i + \tfrac{n_{H,i}}{2}\,\lambda\right). \tag{3}$$

2.3 Mass and volumetric basis

Dividing the molar values by the molar mass gives the mass basis, and dividing by the normal molar volume $V_m = 22.414 \times 10^{-3}\,\mathrm{Nm^3/mol}$ (at $0\,^\circ\mathrm{C}$, $101.325\,\mathrm{kPa}$) gives the volumetric basis and the gas density:

$$h_L = \frac{\mathrm{LHV_{mol}}}{M_f}, \qquad h_H = \frac{\mathrm{HHV_{mol}}}{M_f}, \tag{4}$$

$$\rho_N = \frac{M_f/1000}{V_m}, \qquad \mathrm{LHV_{vol}} = \frac{\mathrm{LHV_{mol}}/1000}{V_m}. \tag{5}$$

Here h_L, h_H come out in MJ/kg, ρ_N in $\mathrm{kg/Nm^3}$, and $\mathrm{LHV_{vol}}$ in $\mathrm{MJ/Nm^3}$.

2.4 Wobbe index

The Wobbe index measures burner interchangeability. With the fuel specific gravity relative to air $s = M_f/28.964$:

$$W = \frac{\mathrm{HHV_{vol}}}{\sqrt{s}}. \tag{6}$$

Two fuels with the same Wobbe index deliver the same heat through a given burner at a given pressure, even if their compositions differ.

2.5 Stoichiometric air

Complete combustion of a component $C_xH_yS_z$ follows

$$C_xH_yS_z + \left(x + \tfrac{y}{4} + z\right) O_2 \longrightarrow x\, CO_2 + \tfrac{y}{2}\, H_2O + z\, SO_2. \tag{7}$$

The oxygen demand per mole of fuel mixture, the air demand, and the stoichiometric air-to-fuel mass ratio are

$$n_{O_2} = \sum_i x_i \left(x_i^C + \tfrac{y_i^H}{4} + z_i^S\right), \qquad \mathrm{AFR} = \frac{n_{O_2}(1+\beta)\cdot 28.964}{M_f}, \tag{8}$$

Table 1. Per-component data. Heating values are standard molar lower heating values; oxygen demand is derived from Eq. (7).

Component	M_i (g/mol)	LHV$_i$ (kJ/mol)	C	H	S	role
H_2	2.016	241.8	0	2	0	fuel
CH_4	16.043	802.3	1	4	0	fuel
C_2H_6	30.070	1428.6	2	6	0	fuel
C_2H_4	28.054	1323.2	2	4	0	fuel
C_3H_8	44.097	2043.9	3	8	0	fuel
C_3H_6	42.081	1926.1	3	6	0	fuel
i-C_4H_{10}	58.123	2649.0	4	10	0	fuel
n-C_4H_{10}	58.123	2657.3	4	10	0	fuel
CO	28.010	283.0	1	0	0	fuel
CO_2	44.010	0	1	0	0	inert
N_2	28.014	0	0	0	0	inert
H_2S	34.081	518.0	0	2	1	fuel, sulfur
H_2O	18.015	0	0	0	0	inert

where x_i^C, y_i^H, z_i^S are the carbon, hydrogen and sulfur atom counts of component i.

For a representative refinery fuel gas (25 % H_2, 45 % CH_4, with C_2–C_4 and inerts) the model returns $M_f \approx 19\,\mathrm{g/mol}$, $h_L \approx 44.5\,\mathrm{MJ/kg}$, $W \approx 51\,\mathrm{MJ/Nm^3}$ and AFR ≈ 15.2, all inside normal ranges.

3 Combustion product balance

Given the fuel and an excess-air fraction e, the flue-gas composition follows from a species balance. Let $n_C = \sum_i x_i x_i^C$ be the total carbon per mole of fuel, and define the water and sulfur products similarly.

3.1 Incomplete combustion

A combustion efficiency $\varepsilon \leq 1$ diverts a carbon fraction $(1-\varepsilon)$ to carbon monoxide rather than carbon dioxide:

$$n_{CO} = n_C\,(1-\varepsilon), \qquad n_{CO_2} = n_C - n_{CO}. \tag{9}$$

Carbon monoxide needs only half the oxygen of full combustion, so the oxygen actually consumed is $n_{O_2}^{\mathrm{cons}} = n_{O_2} - \frac{1}{2} n_{CO}$.

3.2 Excess air and flue gas

Oxygen is supplied at $n_{O_2}^{\mathrm{sup}} = n_{O_2}(1+e)$. The leftover oxygen and the nitrogen brought in with the air are

$$n_{O_2}^{\mathrm{exc}} = n_{O_2}^{\mathrm{sup}} - n_{O_2}^{\mathrm{cons}}, \qquad n_{N_2}^{\mathrm{air}} = \beta\, n_{O_2}^{\mathrm{sup}}. \tag{10}$$

The wet and dry flue-gas totals per mole of fuel are

$$n_{\mathrm{wet}} = n_{CO_2} + n_{H_2O} + n_{SO_2} + n_{CO} + n_{O_2}^{\mathrm{exc}} + n_{N_2}, \qquad n_{\mathrm{dry}} = n_{\mathrm{wet}} - n_{H_2O}. \tag{11}$$

The measured dry-basis concentrations, which the control system reads, are

$$O_2^{\text{dry}} = 100\,\frac{n_{O_2}^{\text{exc}}}{n_{\text{dry}}}, \quad CO_2^{\text{dry}} = 100\,\frac{n_{CO_2}}{n_{\text{dry}}}, \quad CO = 10^6\,\frac{n_{CO}}{n_{\text{dry}}}, \quad SO_2 = 10^6\,\frac{n_{SO_2}}{n_{\text{dry}}}. \tag{12}$$

The stack oxygen O_2^{dry} is the quantity that feeds the efficiency KPI in Section 5.

4 Steady-state heater model

The model resolves one operating instant. The logic is: compute the duty the process demands, model the stack temperature, obtain the efficiency, then back out the fuel needed to meet the duty at that efficiency.

4.1 Heat duty

The heat the process requires is

$$Q = \dot{m}_p\, c_{p,p}\,(T_{\text{out}} - T_{\text{in}}), \tag{13}$$

evaluated in kW with $\dot{m}_p$ in kg/s, then reported in MW and in GJ/h (where $1\,\text{MW} = 3.6\,\text{GJ/h}$). Equation (13) is the first dashboard KPI.

4.2 Firing ratio and stack temperature

A firing ratio compares the current duty to a nominal design duty Q_0: $r = Q/Q_0$. The stack temperature rises with excess air, fouling, ambient temperature and load:

$$T_s = T_s^0 + 220\,(e - e_0) + 55\,\phi + 0.6\,(T_a - T_a^0) + 40\,(r - 1), \tag{14}$$

with design values $T_s^0 = 160\,°\text{C}$, $e_0 = 0.15$, $T_a^0 = 25\,°\text{C}$. The coefficient on e reflects the extra flue-gas mass that must carry the same heat up the stack; the coefficient on ϕ reflects lost convection recovery as the tubes foul.

4.3 Flame temperature

An adiabatic first-law estimate divides the released heat by the heat capacity of the products. With mass of products per unit fuel mass $m_{\text{pr}} = 1 + \text{AFR}(1 + e)$ and a mean hot flue specific heat $c_{p,g} = 1.35\,\text{kJ/(kg K)}$:

$$T_{\text{flame}} = T_a + \frac{h_L \cdot 1000}{m_{\text{pr}}\, c_{p,g}}. \tag{15}$$

For the reference fuel this gives about 1805 °C, consistent with a gas flame at moderate excess air.

4.4 Thermal nitrogen oxides

Thermal nitrogen oxide formation rises steeply with flame temperature (a Zeldovich-type dependence) and with available oxygen. With T_K the flame temperature in kelvin:

$$\text{NO}_\text{x} = \max\left(5,\ 41\,\exp\frac{T_K - 2078}{150}\,(1 + 3e)\right)\ \text{ppm}. \tag{16}$$

The constants are set so the reference case gives about 60 ppm.

4.5 Efficiency (indirect method)

Efficiency is the workbook correlation, an indirect stack-loss method. With the stack oxygen $O_2 \equiv O_2^{\text{dry}}$ and stack temperature T_s:

$$\boxed{\eta = 100 - 2.5 - \left[\left(0.044 + 0.0325\,\frac{O_2}{18.16 - O_2}\right)(T_s - 30) - 0.8\right]} \tag{17}$$

The fixed 2.5 % term is an unaccounted-loss allowance and the bracket is the dry-gas and moisture loss. Equation (17) reproduces the workbook value of 92.397 % at $O_2 = 2.5\,\%$, $T_s = 150\,°\text{C}$. The result is clamped to $[60, 95]\,\%$ for physical safety.

4.6 Fuel firing and combustion air

The fuel must supply the duty at the computed efficiency, so the fuel energy and mass flow are

$$\dot{E}_f = \frac{Q}{\eta/100} \quad [\text{GJ/h}], \qquad \dot{m}_f = \frac{\dot{E}_f \cdot 10^6}{h_L \cdot 1000} \quad [\text{kg/h}]. \tag{18}$$

Equation (18) closes the energy balance exactly: $\dot{E}_f\,(\eta/100) = Q$ to machine precision. The stoichiometric and actual combustion air are

$$\dot{m}_{\text{air}}^{\text{stoich}} = \dot{m}_f\,\text{AFR}, \qquad \dot{m}_{\text{air}} = \dot{m}_{\text{air}}^{\text{stoich}}(1 + e). \tag{19}$$

4.7 Heat split and tube metal temperature

The radiant section takes a fixed fraction $f_r = 0.70$ of the duty and the convection section takes the rest. The tube metal (skin) temperature is modelled as the process outlet plus a radiant driving term and a fouling penalty:

$$Q_r = f_r Q, \qquad Q_c = (1 - f_r)Q, \qquad T_{\text{tube}} = T_{\text{out}} + 90 + 60\,r + 40\,\phi. \tag{20}$$

The tube metal temperature is a safety constraint, since exceeding the metallurgical limit shortens tube life.

4.8 Emissions

Carbon dioxide is reported with the workbook emission-factor method, using the factor $F = 56.1\,\text{kg/GJ}$ of fuel:

$$\dot{m}_{CO_2} = \frac{\dot{E}_f\,F}{1000} \quad [\text{t/h}]. \tag{21}$$

Sulfur oxides follow from the fuel sulfur through the combustion balance of Section 3.

5 The four dashboard KPIs

The source workbook defines four target KPIs. Table 2 maps each to the equation above.

6 Time-series data generation

The generator drives the steady-state model through an operating history and adds measurement noise and labelled faults, producing a historian-style dataset. The default set is 120 days at hourly resolution, giving 2,880 records.

Table 2. Dashboard KPIs and their governing equations.

KPI	Equation	Notes
Heat duty	Eq. (13)	$Q = \dot{m}_p c_{p,p} \Delta T$
Efficiency	Eq. (17)	indirect stack-loss method
CO_2 emission	Eq. (21)	fuel energy times 56.1 kg/GJ
Heater losses	$L_s = 100 - \eta - 2.5$, $L_{\text{tot}} = L_s + 2 + 2.5$	stack plus refractory (2 %) plus unaccounted

6.1 Operating drivers

Slow drivers use a mean-reverting (Ornstein–Uhlenbeck) process,

$$X_t = X_{t-1} + \theta\,(X_0 - X_{t-1}) + \sigma\,\xi_t, \qquad \xi_t \sim \mathcal{N}(0,1), \tag{22}$$

which drifts around a set point X_0 without wandering off. Throughput adds occasional step changes for campaign turndowns. Ambient temperature combines a seasonal and a daily cycle,

$$T_a = 25 + 8\sin\big(2\pi(\tau - 0.3)\big) + 5\sin\Big(\tfrac{2\pi(h-6)}{24}\Big) + \varepsilon, \tag{23}$$

with τ the fraction of the year and h the hour of day. Fouling grows over a campaign and resets at a cleaning event, giving a sawtooth in ϕ. The fuel hydrogen content drifts between 12 and 45 mol%, with methane taking the balance, so the fuel properties change over time as they do on a real unit.

6.2 Labelled faults

Seven fault classes are injected, each over a random window, and every row carries a flag and a class label. Process faults are real physical excursions the twin must learn; sensor faults corrupt only the measurement and should be removed before training. Table 3 lists them.

Table 3. Injected fault classes and their signatures.

Class	Kind	Signature
Air deficiency	process	carbon monoxide breakthrough, oxygen falls
High excess air	process	efficiency falls, stack loss rises
Fouling spike	process	stack temperature rises, efficiency falls
Fuel upset	process	heavy-gas slug, heating value falls
Oxygen sensor stuck	sensor	stack oxygen reading frozen
Flow spike	sensor	flow transmitter transient
Stack-temperature bias	sensor	stack temperature offset

6.3 Measurement noise

Each historian tag receives Gaussian noise, proportional for flows and duty (about one percent) and absolute for temperatures and concentrations. Non-negative quantities such as carbon monoxide are floored at zero.

7 The machine-learning twin

7.1 Clean and label

Cleaning removes the sensor-fault rows, since their measurements are corrupt. Process-fault rows are kept because they are legitimate operating states. Of the 2,880 raw rows, 105 sensor-fault rows are removed, leaving 2,775 clean rows, of which 126 are real process-fault states.

7.2 Features and targets

The twin learns the heater response from the controllable and disturbance inputs. The nine input features are process flow, inlet and outlet temperature, process pressure, excess air, fuel hydrogen fraction, fuel lower heating value, fouling index and ambient temperature. The six targets are efficiency, stack temperature, stack oxygen, carbon dioxide, tube metal temperature and fuel firing.

7.3 Model and scoring

Each target is fitted with a random forest of 250 trees on a random 75 % train split, and scored on the held-out 25 %. The scores are the coefficient of determination R^2, the mean absolute error, and the mean absolute percentage error. Table 4 gives the results.

Table 4. Twin accuracy on the hold-out test set (694 rows).

Target	R^2	MAE	MAPE (%)
Efficiency (%)	0.992	0.052	0.06
Carbon dioxide (t/h)	0.990	0.058	0.58
Stack temperature (°C)	0.982	1.553	0.89
Stack oxygen (%)	0.979	0.045	1.55
Tube metal temp (°C)	0.967	2.655	0.48
Fuel firing (kg/h)	0.960	47.30	1.21

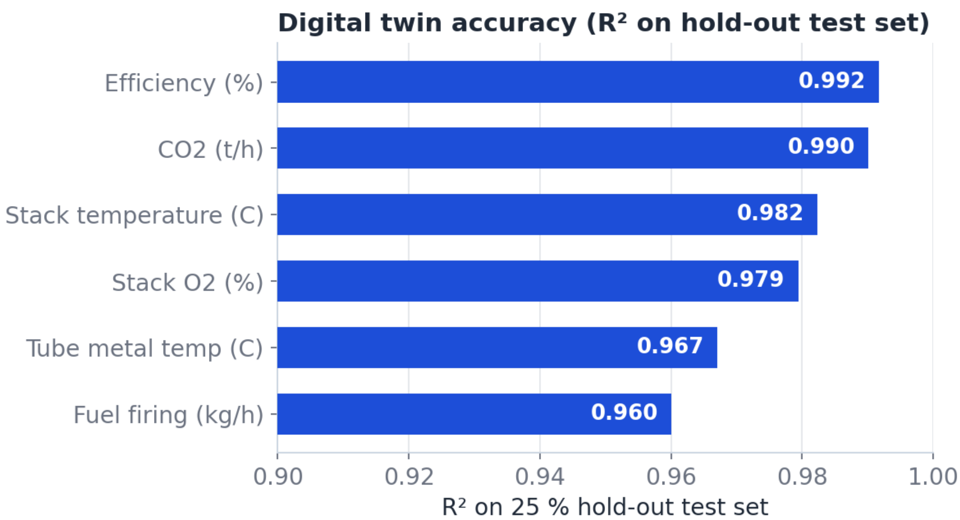

Figure 1. Twin accuracy per target, all between 0.96 and 0.99, against the first-principles truth.

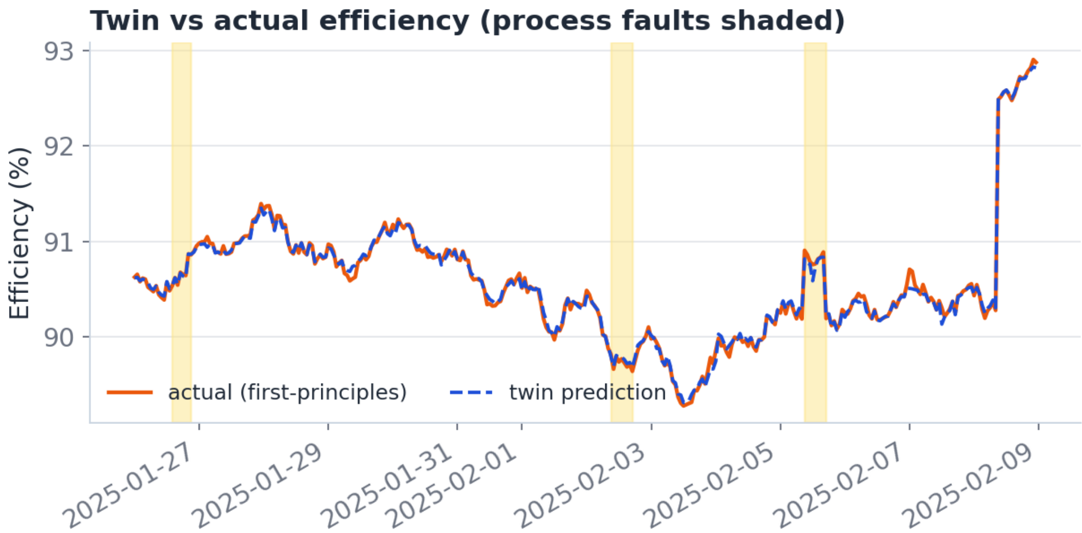

Figure 2. Predicted efficiency traces the actual efficiency across a fourteen-day window that includes drift and shaded process faults.

8 Operating-point optimization

The value layer finds the excess-air setpoint that minimises fuel, and therefore carbon, without crossing the combustion or corrosion limits.

8.1 Carbon monoxide against excess air

Below a knee, cutting air lets carbon escape as carbon monoxide. The unburned fraction is modelled as

$$1 - \varepsilon(e) = \min\Big(0.03\,\exp\big(-(e-0.03)/0.025\big),\ 0.05\Big), \tag{24}$$

which gives negligible carbon monoxide above about 12 % air and a sharp rise below about 8 %. This creates a genuine optimum: too much air wastes heat up the stack, too little makes carbon monoxide.

8.2 Objective and constraints

For a fixed duty and fuel, the optimizer sweeps the excess air and selects the feasible point that minimises the fuel energy $\dot{E}_f$:

$$\min_{e}\ \dot{E}_f(e) \quad \text{s.t.} \quad CO(e) \le 200\,\text{ppm}, \quad T_s(e) \ge 130\,^{\circ}\text{C}, \quad T_{\text{tube}}(e) \le 620\,^{\circ}\text{C}. \tag{25}$$

The lower stack-temperature bound is the acid dew point, below which the stack corrodes.

8.3 Savings and economics

Comparing the current operation with the optimum gives the savings. With fuel price $p_f = 6\,\text{USD/GJ}$, carbon price $p_c = 40\,\text{USD/t}$ and $H = 8400$ operating hours per year:

$$\Delta\dot{E}_f = \dot{E}_f^{\text{base}} - \dot{E}_f^{\text{opt}}, \qquad \Delta\dot{m}_{CO_2} = \dot{m}_{CO_2}^{\text{base}} - \dot{m}_{CO_2}^{\text{opt}}, \tag{26}$$

$$\text{Fuel value} = \Delta\dot{E}_f\,H\,p_f, \qquad \text{Carbon value} = \Delta\dot{m}_{CO_2}\,H\,p_c. \tag{27}$$

8.4 Result

For a representative point currently run at 24 % excess air, the optimizer recommends 10 %, where carbon monoxide just reaches its limit. Table 5 summarises the outcome.

Table 5. Optimization result for one heater at the stated prices.

Quantity	Current	Recommended
Excess air (%)	24.0	10.0
Stack oxygen (%)	4.41	2.10
Efficiency (%)	89.40	91.89

Efficiency gain: **+2.49 points**
Fuel saved: 2.71 % (5.43 GJ/h)
Carbon avoided: 2560 t/yr
Combined value: **about** 376 000 USD **per year, one heater**

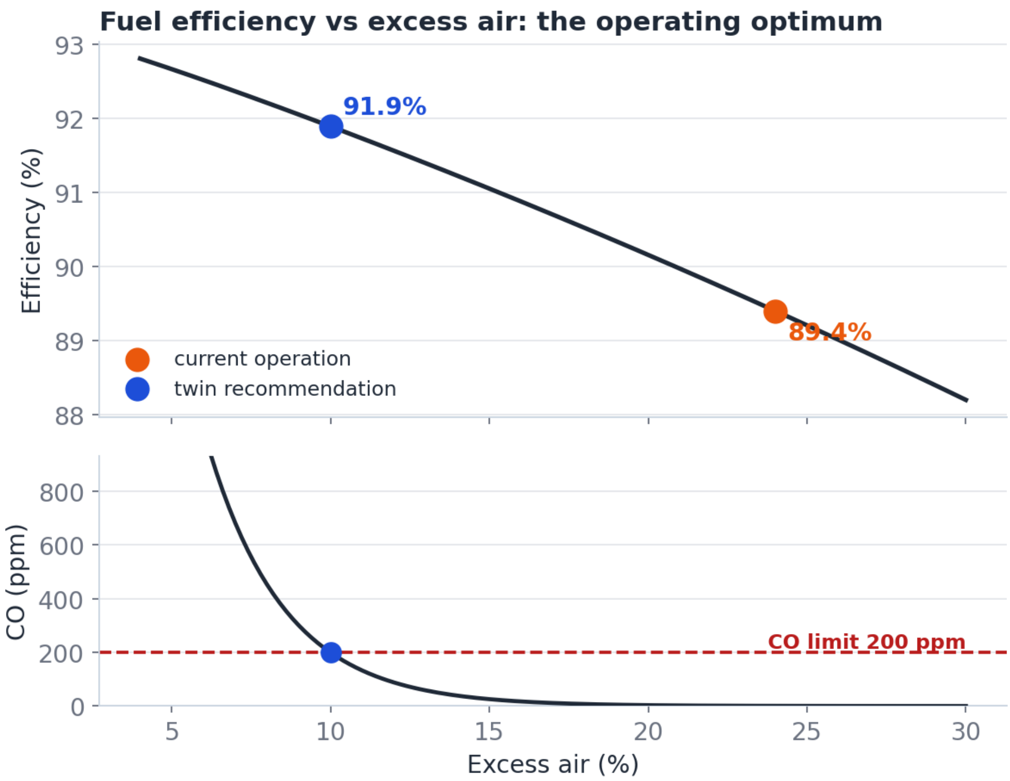

Figure 3. Efficiency against excess air (top) and carbon monoxide against excess air (bottom). The optimum sits just above the carbon monoxide limit. Orange marks the current operation, blue the recommendation.

Constant	Symbol	Value
Process specific heat	$c_{p,p}$	2.55 kJ/(kg K)
Design stack temperature	T_s^0	160 °C
Design excess air	e_0	15 %
Radiant fraction	f_r	0.70
Hot flue specific heat	$c_{p,g}$	1.35 kJ/(kg K)
Carbon dioxide factor	F	56.1 kg/GJ
Refractory loss		2 %
Unaccounted loss		2.5 %
Fuel price	p_f	6 USD/GJ
Carbon price	p_c	40 USD/t
Operating hours	H	8400 per year

File	Role
`fuel_properties.py`	Sections 2–3: fuel and combustion
`heater_model.py`	Sections 4–5: energy balance and KPIs
`generate_dataset.py`	Section 6: dataset generation
`twin_pipeline.py`	Section 7: clean, label, train, validate
`optimize.py`	Section 8: optimizer
`make_charts.py`	the figures in this report
`dashboard.py`	Streamlit live demo

9 Reproducibility

9.1 Constants

9.2 Code map and run order

Run order: `generate_dataset.py` → `twin_pipeline.py` → `optimize.py` → `make_charts.py` → `streamlit run dashboard.py`.

10 Assumptions and limitations

- The results come from a first-principles simulation, not a live plant. The reported accuracy is the surrogate against the physics model, not against field measurements. The next milestone is calibration on a real heater.
- The steady-state model resolves one instant at a time and does not capture fast transients or detailed burner aerodynamics.
- The flame temperature and nitrogen-oxide relations are calibrated estimates, adequate for trend and optimization, not for detailed emissions design.
- The economics use indicative prices. The physical savings, fuel and carbon per hour, do not depend on those prices.